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United States Patent	12408807
Kind Code	B2
Date of Patent	September 09, 2025
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Structure for applying antimicrobial material to a cable

Abstract

A puck is made of a porous resilient material and is held in a housing to comprise a cable swipe cassette. The puck holds fluid in the porous material, and there is a bore through the puck. A cable is routed through the bore so that when the cable is pulled through the bore the exterior surface of the cable is wetted with the fluid. The fluid can have an antimicrobial effect so as to disinfect the cable.

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Family ID:	93293689
Appl. No.:	18/655785
Filed:	May 06, 2024

Prior Publication Data

Document Identifier	Publication Date
US 20240366822 A1	Nov. 07, 2024

Related U.S. Application Data

us-provisional-application US 63464323 20230505

Publication Classification

Int. Cl.: A47L13/17 (20060101); A61B50/24 (20160101); A61B90/70 (20160101); A61L2/235 (20060101); B05C1/06 (20060101)

U.S. Cl.:

CPC **A47L13/17** (20130101); **A61B90/70** (20160201); **A61L2/235** (20130101); **B05C1/06** (20130101); A61B50/24 (20160201); A61L2202/15 (20130101); A61L2202/24 (20130101)

Field of Classification Search

CPC: A61B (90/70); A61L (2202/24); A61L (2/235); A47L (13/17); A47L (13/26); A47L (25/00); B05C (1/06)

USPC: 15/104.04; 15/104.92-1

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Background/Summary

CROSS REFERENCE (1) This application claims priority to U.S. provisional application No. 63/464,323, filed May 5, 2023, the entirety of which is hereby incorporated by reference.

FIELD OF THE INVENTION

(1) The present invention relates generally to medical equipment, and, more particularly, relates to a structure or device that applies antimicrobial material to cables used for medical devices, and specifically to a device that holds a puck or similar article that has a hole through it, wherein the medical device cable is routed through the hole in the puck. The puck contains antimicrobial material, and when the cable is pulled through the puck, such as for use in measuring a biometric parameter, some of the antimicrobial material is transferred to the outer surface of the cable, thereby killing microbes on the outer surface of the cable.

BACKGROUND OF THE INVENTION

(2) It is well known that in the provision of medical services, it is imperative to prevent exposing a patient to infectious microbes. There are numerous surfaces and instruments that patients come in contact with in a medical facility that can impart undesirable microbes to the patient through contact with those items. While routine cleaning practices can reduce the incidence of infectious microbes, some surfaces, like floors, are frequently being contaminated simply through ordinary usage (e.g. people walking on them with dirty shoes). Another surface that is easily contaminated, and often overlooked for cleaning, are the cables used with medical instruments. These are often overlooked, and can come into contact with various other surfaces, including floors. Given how often medical instruments are used, and moved from patient to patient, they present a likely infection vector. Cables are often placed on the patient during use, and therefore come into contact with the garments worn by the patient, who also may make contact with any material left on the garment from the cables. In some cases, the patient can come into direct contact with the cables. Even if carefully kept from coming into contact with potentially unclean surfaces, such as a floor, the cables can come into contact with multiple patients, who may have an infection or unknowingly have infectious material on their clothing. Thus, cables of medical devices present a significant potential for transmitting infectious agents. Cable can be cleaned by personnel by, for example, spraying the cables with a disinfecting agent and then wiping them down, but this adds to the tasks needed to be performed by personnel, and in practice, it is often foregone as a matter of convenience.

(3) Therefore, a need exists to overcome the problems with the prior art as discussed above.

SUMMARY

(4) In accordance with some embodiments of the inventive disclosure, there is provided a cable swipe cassette which includes a puck made of a porous resilient material. The puck has a bore from a first side to a second side opposite the first side. There is also a housing that encloses the puck therein. The housing has openings on opposite sides of the housing that correspond with the bore.

(5) In accordance with a further feature, the housing is hinged and has an upper portion and a lower portion coupled together at a hinge.

(6) In accordance with a further feature, the housing has a latch opposite the hinge to latch the upper and lower portions of the housing together.

(7) In accordance with a further feature, there is also included a fluid disposed in the puck.

(8) In accordance with a further feature, the fluid is an antimicrobial fluid.

(9) In accordance with a further feature, the puck further comprises a slot extending outward from the bore to a cylindrical surface of the puck.

(10) In accordance with some embodiments of the inventive disclosure, there is provided a stand for a medical device that includes a cable retracting mechanism disposed on the stand. There is a cable disposed on the cable retracting mechanism that has a first end that is configured to attach to a signal input of the medical device and a second end that is configured to attach to a sensor. There is also a cable swipe that includes a holder that can be opened. The holder has opposing walls, and each of the opposing walls has an opening through them. There is a puck configured to be disposed in the holder. The puck made of a material configured to hold an antimicrobial fluid and has a bore through it that, when the puck is placed in the holder, aligns with the openings in the opposing walls of the holder. The bore through the puck has a smaller diameter than an exterior diameter of the cable. The puck further has a slot from a cylindrical surface of the puck to the bore through the puck. The second end of the cable passes through the bore of the puck such that when the cable is drawn through the puck an exterior of the cable is wetted by the antimicrobial fluid. There is also a support mounted on the stand that is configured to support the holder while the holder is holding the puck and while the cable passes through the puck.

(11) In accordance with a further feature, the puck is made of a porous resilient material.

(12) In accordance with a further feature, the puck is selected from a plurality of pucks that have different sized bores.

(13) In accordance with a further feature, the stand comprises casters in order to make the stand mobile.

(14) In accordance with some embodiments of the inventive disclosure, there is provided a cable disinfecting mechanism that includes a cable reel. The cable reel has a cable wound on a reel. The cable reel has a cable outlet through which one end of the cable passes. There is also a cable swipe cassette assembly mounted at the cable outlet. The cable swipe cassette assembly has a housing and a puck inside the housing. The puck is made of a resilient porous material and has a bore through it. There is an antimicrobial fluid disposed in the puck. The one end of the cable passes through the bore of the puck such that when the cable is drawn through the puck an exterior surface of the cable is wetted with the antimicrobial fluid.

(15) In accordance with a further feature, the puck includes a slot that passes through a side of the puck to the bore.

(16) In accordance with a further feature, the housing of the cable swipe cassette assembly is configured to clamp onto the cable reel at the cable outlet.

(17) In accordance with a further feature, the cable reel is mounted on a stand for a medical device.

(18) In accordance with a further feature, the housing of the cable swipe cassette assembly is hinged and has an upper portion and a lower portion coupled together at a hinge.

(19) In accordance with a further feature, the housing has a latch opposite the hinge to latch the upper and lower portions of the housing together.

(20) Although the invention is illustrated and described herein as embodied in a structure for applying antimicrobial material to cables for medical devices, it is, nevertheless, not intended to be limited to the details shown because various modifications and structural changes may be made therein without departing from the spirit of the invention and within the scope and range of equivalents of the claims. Additionally, well-known elements of exemplary embodiments of the invention will not be described in detail or will be omitted so as not to obscure the relevant details of the invention.

(21) Other features that are considered as characteristic for the invention are set forth in the appended claims. As required, detailed embodiments of the present invention are disclosed herein; however, it is to be understood that the disclosed embodiments are merely exemplary of the invention, which can be embodied in various forms. Therefore, specific structural and functional details disclosed herein are not to be interpreted as limiting, but merely as a basis for the claims and as a representative basis for teaching one of ordinary skill in the art to variously employ the present invention in virtually any appropriately detailed structure. Further, the terms and phrases used herein are not intended to be limiting; but rather, to provide an understandable description of the invention. While the specification concludes with claims defining the features of the invention that are regarded as novel, it is believed that the invention will be better understood from a consideration of the following description in conjunction with the drawing figures, in which like reference numerals are carried forward. The figures of the drawings are not drawn to scale.

(22) Before the present invention is disclosed and described, it is to be understood that the terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting. The terms “a” or “an,” as used herein, are defined as one or more than one. The term “plurality,” as used herein, is defined as two or more than two. The term “another,” as used herein, is defined as at least a second or more. The terms “including” and/or “having,” as used herein, are defined as comprising (i.e., open language). The term “coupled,” as used herein, is defined as connected, although not necessarily directly, and not necessarily mechanically. The term “providing” is defined herein in its broadest sense, e.g., bringing/coming into physical existence, making available, and/or supplying to someone or something, in whole or in multiple parts at once or over a period of time.

(23) “In the description of the embodiments of the present invention, unless otherwise specified, azimuth or positional relationships indicated by terms such as “up”, “down”, “left”, “right”,

“inside”, “outside”, “front”, “back”, “head”, “tail” and so on, are azimuth or positional relationships based on the drawings, which are only to facilitate description of the embodiments of the present invention and simplify the description, but not to indicate or imply that the devices or components must have a specific azimuth, or be constructed or operated in the specific azimuth, which thus cannot be understood as a limitation to the embodiments of the present invention. Furthermore, terms such as “first”, “second”, “third” and so on are only used for descriptive purposes, and cannot be construed as indicating or implying relative importance.

(24) In the description of the embodiments of the present invention, it should be noted that, unless otherwise clearly defined and limited, terms such as “installed”, “coupled”, “connected” should be broadly interpreted, for example, it may be fixedly connected, or may be detachably connected, or integrally connected; it may be mechanically connected, or may be electrically connected; it may be directly connected, or may be indirectly connected via an intermediate medium. As used herein, the terms “about” or “approximately” apply to all numeric values, whether or not explicitly indicated. These terms generally refer to a range of numbers that one of skill in the art would consider equivalent to the recited values (i.e., having the same function or result). In many instances these terms may include numbers that are rounded to the nearest significant figure. Those skilled in the art can understand the specific meanings of the above-mentioned terms in the embodiments of the present invention according to the specific circumstances.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The accompanying figures, where like reference numerals refer to identical or functionally similar elements throughout the separate views and which together with the detailed description below are incorporated in and form part of the specification, serve to further illustrate various embodiments and explain various principles and advantages all in accordance with the present invention.

(2) FIG. 1 shows a medical device on a stand for use in a medical facility wherein the stand includes a disinfecting cable swipe through which the cable is fed so that every time a cable is drawn through the cable swipe a new application of antimicrobial material is applied to the exterior of the cable.

(3) FIG. 2 shows an alternate example of a medical device on a stand for use in a medical facility wherein the stand includes a disinfecting cable swipe through which the cable is fed so that every time a cable is drawn through a cable swipe a new application of antimicrobial material is applied to the exterior of the cable.

(4) FIG. 3 shows another alternative example medical device on a stand for use in a medical facility wherein the stand includes a disinfecting cable swipe through which the cable is fed so that every time a cable is drawn through a cable swipe a new application of antimicrobial material is applied to the exterior of the cable.

(5) FIG. 4 shows a cable swipe cassette assembly, in accordance with some embodiments.

(6) FIG. 5 shows an alternative cable swipe cassette assembly having a latch, in an open configuration, in accordance with some embodiments.

(7) FIG. 6 shows an alternative cable swipe cassette assembly having a latch, in a closed configuration, in accordance with some embodiments.

(8) FIG. 7 shows a cable reel housing for use with a cable swipe cassette assembly, in accordance with some embodiments.

(9) FIG. 8 shows an exploded view of a cable reel for using with a cable swipe cassette assembly, in accordance with some embodiments.

(10) FIG. 9 shows a medical device stand using several cable reels, each with a cable swipe

cassette, in accordance with some embodiments.

(11) FIG. **10** shows a side view of a cable reel and a clamp on cable swipe cassette, separated, in accordance with some embodiments.

(12) FIG. **11** shows a side view of a cable reel and a clamp on cable swipe cassette, with the cable swipe cassette clamped onto the cable reel, in accordance with some embodiments.

(13) FIG. **12** shows a series of several cable swipes that all fit in the same cassette but which accommodate cables of different sizes, in accordance with some embodiments.

DETAILED DESCRIPTION

(14) While the specification concludes with claims defining the features of the invention that are regarded as novel, it is believed that the invention will be better understood from a consideration of the following description in conjunction with the drawing figures, in which like reference numerals are carried forward. It is to be understood that the disclosed embodiments are merely exemplary of the invention, which can be embodied in various forms.

(15) To solve the problem of medical device cables becoming fouled with potentially infectious agents, a cable swipe is provided that includes a load of antimicrobial material. The cable is drawn through the cable swipe, and contact between the cable and the cable swipe material results in the antimicrobial material wetting the outer surface of the cable along the length of the cable that is drawn through the cable swipe, thereby disinfecting that portion of the cable. Furthermore, in conjunction with the pull-through cable swipe structure, the cable can be provided on a reel so that when the cable is reeled in after use, the cable will again be wetted with the antimicrobial material, and the wetted portion of the cable will be in contact with some portions of the cable that were not drawn through the cable swipe, thereby imparting antimicrobial material to those portions of the cable as well.

(16) FIG. **1** shows a medical device **100** on a stand **102** for use in a medical facility, wherein the stand **102** that holds the medical device **100** can include casters, wheels, or rollers at the bottom of the stand **102** to make the stand **102** and medical device **100** portable so that it can be moved from patient to patient. That is, the medical device **100** can be moved from patient to patient, as needed, to either provide some medical treatment to the patient or to measure and monitor some physiological and medically significant parameter of the patient. The medical device **100** uses one or more cables such as cable **104** to connect between the medical device **100** and a sensor or actuator at or on the patient. Electrical signals are transmitted through the cable **104** either by or to the medical device **100**. Cable **104** is shown here on a cable retracting mechanism such as, for example reel **105** that is held by a reel holder **107** that can mount on the stand **102**. The cable **104** has a portion that is wound around the reel **105**, and a portion that is static which connects to the medical device **100**. An end of the cable **104** is fed through a disinfecting cable swipe **106** that is mounted at the outlet of the reel **105** on the reel holder **107**. The cable **104** is fed through the cable swipe **106** so that every time the cable **104** is drawn through the cable swipe **106** a new application of antimicrobial material is applied to the exterior of the cable **104**. The reel **105** can be self-reeling so that when the medical device **100** is no longer needed for a given patient, the reel **105** can self-wind the cable **104** back onto the reel **105**. The cable swipe **106** is a cassette that includes a puck that can be made of a resilient porous material that is saturated with an antimicrobial fluid. The cable swipe **106**, including the puck held in a case or housing, can be mounted at the outlet of the reel **105** on the reel holder **107**. The reel holder **107** can be attached to the stand **102** by any conventional attachment means, including a bolted bracket, a slip-on attachment, straps, and so on. In some cases, the reel **105** and reel holder **107** can be permanently mounted on the stand **102**.

(17) As a result of being moved from patient to patient, the cable **104** can come into contact with various objects, including a patient, the floor, the hands of various nurses and other healthcare workers, and so on. As a result, there is a significant chance of some microbial agent being introduced to the outside surface of the cable **104**. By drawing the cable **104** through the cable swipe **106** each time it is to be used, and each time it is reeled back up, the outside of the cable **104**

is repeatedly treated with the antimicrobial material. There are a wide variety of disinfecting materials that are known and which can be carried and applied by the cable swipe **106**.

(18) As mentioned, one end of the cable **104** extends out of the reel **105** and attaches to a signal input of the medical device **100**. The other end of the cable **104** is intended to be attached to a sensor on a patient, and is fed off the reel **105** through the cable swipe **106**, which is held in a cable swipe support **108** that can be mounted on the reel holder **107**. The cable swipe **106** can include, for example a puck or an annular sponge member that has an opening **109** through which the cable **104** passes. The opening **109** is sized to be slightly smaller than the exterior diameter of the cable **104** to ensure that the cable swipe **106** will be in contact with the exterior of the cable **104** so as to impart the antimicrobial material to the outside of the cable **104**. The cable swipe **106** can be made of a porous resilient material that contains or holds the antimicrobial fluid.

(19) FIG. 2 shows an alternate example of a medical device **202** on a stand **204** for use in a medical facility wherein the stand **204** includes a disinfecting cable swipe **210** through which a cable **212** is fed so that every time a cable is drawn through a cable swipe **210** a new application of antimicrobial material is applied to the exterior of the cable **212**. The stand **204** can include a wheeled base **206** so as to be mobile. A reel or other cable retracting mechanism **214** can be contained in or on the stand **204** that the cable **212** can be locked in position once the cable **212** is fed out to a desired length through a guide **208**, and then unlocked to reel in the cable **212**. The guides **208** are configured to support the puck in the cable swipe as well as guide the cable out of and into the cable retracting mechanism. The guides can be, for example, tubular arms that extend from a central post of the stand **204**. The cable retracting mechanism **214** can include a pulley **216** around which the cable **212** is routed, and as the cable **212** is pulled out, through the cable swipe **210**, the pulley **216** rises inside or on the outside of the stand **204**, and may be biased downward by a spring, or simply its own weight. The position of the pulley **216** can be locked in place once the desired length of cable **212** is fed out, and then released to draw the cable back into the stand **204**. When the cable **212** is drawn back into the stand **204** by the pulley **216** being urged downward, more of the antimicrobial fluid is swiped onto the exterior surface of the cable **212**. Thus, as the cable **212** travels in both directions (being fed out, and then reeled in) it passes through the cable swipe **210**, which is substantially similar to cable swipe **106**. The cable swipe **210** can be changed out periodically to ensure that there is sufficient wetting of the exterior of the cable **212** with the antimicrobial fluid from the cable swipe **210**.

(20) FIG. 3 shows another alternative example medical device system **300** in which a medical device **301** is positioned on a stand **302** for use in a medical facility wherein the stand **302** includes a disinfecting cable swipe **304** through which a cable **308** is fed from a reel **306** so that every time the cable **308** is drawn through the cable swipe **304** a new application of antimicrobial material is applied to the exterior of the cable **308**. Similar to FIG. 2, the medical device stand **302** supports or otherwise includes reel **306**. The reel **306** can be spring driven to automatically take up slack in the cable **308**, or it can be manually driven. An internal end of the cable **308** is attached to a signal input of the medical device **301**, and the other end is fed out through the cable swipe **304** and can be attached to a sensor on the patient. In this embodiment the reel **306** can be housed internally in the stand **302** or externally as in FIG. 1. A in the previous examples, the stand **302** is mobile by having a plurality of casters or wheels at the bottom of the stand **302** on which the stand **302** can roll about.

(21) FIG. 4 shows a cable swipe cassette assembly **400**, in accordance with some embodiments. The assembly **400** can include a holder **402** that is configured to hold a puck **404**. The puck **404** is a portion of material that can hold a volume of antimicrobial fluid. The puck material can be, for example, fabric, open cell resilient foam, or any other suitably resilient and porous material. In general, the material of the puck **404** can be such that capillary action between the antimicrobial fluid and the material of the puck helps to retain the fluid in the puck **404**. The puck **404** has an opening **406** through it that aligns with similar openings in the sides of the holder **402**. The puck

404 has a cut **408** from the outside of the puck **404** to the opening **406** that allows a user to push a cable through so that the cable sits in the opening **406**. As described, the opening **406** is sized so that there is constant contact between the material of the puck **404** and the cable as the cable is drawn back and forth through the opening **406**. Thus, as indicated in FIG. 12, there can be a variety of pucks **404** that all have different sized openings **406** to accommodate different sizes of cables. The holder **402** ensures that the fluid held in the puck **404** doesn't drip out, as well as providing mechanical support for the puck **404**. The holder **402** can be mounted in or on another structure on the stand holding the medical device. The puck **404** can be individually packaged **410** so that, when a puck is depleted of the microbial fluid, it can be easily replaced with a new puck. To replace the puck, the holder **402** is opened, the old puck removed from both the holder **402** and the cable, then the new puck is removed from the packaging **410** and placed in the holder, and the cable is placed into the opening **406** of the new puck by passing the cable through the cut **408**, and then the holder **402** can be closed. The holder **402** can be made of a rigid, fluid impermeable material, and is supported by structure on the stand, such as a tube, guide, reel, or other mounting apparatus that is configured to hold the holder **402**, puck **404**, and cable for use.

(22) FIG. 5 shows an alternative cable swipe cassette assembly **500** having a latch **512**, in an open configuration, and FIG. 6 shows the same assembly **500** in a closed configuration, in accordance with some embodiments. The assembly **500** includes a puck housing or holder **502** and the puck **504** which sits inside the holder **502**. The holder **502** can be a “clamshell” type container that has an upper portion **506** and a lower portion **508** that are joined by a hinge **510** so that the holder encloses the puck **504**. The holder **502** is somewhat puck-shaped itself, and upper and lower portions **506**, **508** can each have a semi cylindrical shape to accommodate the puck **504**, which has a cylindrical shape, inside the holder **502**. The upper portion **506** has a first side **524** and a second side **528**, and the lower portion **508** likewise has a first side **526** and a second side **530**. The first sides **524**, **526** could be considered a front side of the housing, and the second sides **528**, **530** could be considered a back side of the housing when the housing portions **506**, **508** are closed together. The upper portion **506** includes a transverse wall **536** between the first side **524** and the second side **528** that forms half of the outside circumference of the holder **502**. Likewise, the lower portion **508** has a transverse wall **538** between the first side **526** and the second side **530** of the lower portion **508** that forms the complementary half of the outside circumference of the holder **502** when the upper and lower portions **506**, **508** are closed together. The puck **504** sits in the housing **502** between the first sides **524**, **526** and the second sides **528**, **530**, and surrounded around cylindrical side **522** by the transverse walls **536**, **538** of the upper and lower portions **506**, **508**. That is, the puck **504** occupies a space between the first side **524** and the second side **528** of the upper portion **506**, and between the first side **526** and the second side **530** of the lower portion **508** of the housing **502** and within the transverse walls **536**, **538**. Thus, when placed into the housing/holder **502**, a first side **532** of the puck **504**, which is one of the two flat surfaces **532**, **534** of the puck **504**, will be adjacent the first side **524** of the upper portion **506** and the first side **526** of the lower portion **508** inside the holder **502**. Likewise, the second side **534** of the puck **504** will be adjacent the second side **528** of the upper portion **506** and the second side **530** of the lower portion **508** while inside the holder/housing **502**. There can be a latch member **512** that interfaces with a retention feature **514** to hold the holder closed, but allow a user to open the holder by disengaging the latch **512** from the retention member **514**. The latch is on the lower portion **508**, opposite the hinge **510**, and extends upward so as to be able to engage the retention member **514**, which is also opposite the hinge on the upper portion **506**. When the holder **502** is closed, the upper end of the latch member **512** is pushed outward by the retention member **514** as the retention member **514** moves past a retaining lip **515**. Once the retaining member **514** is past the retaining lip **515** the retaining lip **515** will move back toward the center of the holder, thereby creating a mechanical interference between the retaining member **514** and the retaining lip **515**, holding the holder portions **506**, **508** closed. When a user wants to open the holder **502**, to insert a puck **504**, the upper end of the latch member

512 can be biased outward by the user's fingers, allowing the retaining member **514** to move past the retaining lip **515** in the upward or opening direction. The holder **502** has openings on opposite sides or opposite/opposing walls that correspond to the location of the bore through the puck **504**. (23) The puck **504** is made of a resilient porous material that can hold an antimicrobial fluid. Furthermore, the puck **504** has a bore **520** through the puck in the axial direction of the puck **504**. That is, the puck has a cylindrical shape, and the axial direction is from one end to the opposite end of the cylinder-shaped body. There is also a slot **508** cut into the puck from the outside along the side of the puck **504** to the bore **520**. This allows a user to press or push a cable through the slot **518** into the bore **520**, thereby capturing the cable in the bore **520**. With the cable captured in the bore, the puck **504** can then be placed into the holder **502**, and the holder **502** then closed. Thereafter, when the cable is drawn through the puck, in either direction, some of the antimicrobial fluid carried by the puck can be imparted to the cable. The cable will pass through the openings **540** on the first and second sides of the holder **502**. The openings **540** can be circular and formed when the holder/housing **502** is closed by semicircular cutouts **516** formed along the edges of the first sides **524**, **526** where the first side **524** of the upper portion **506** meets the first side **526** of the lower portion **508** when the holder **502** is closed, and at the edges of the second sides **528**, **530** where the second side **528** of the upper portion **506** meets the second side **530** of the lower portion **508** when the holder **502** is closed. The holder **502**, when closed, can slightly compress the puck **504** so as to prevent the cable from becoming seated in the slot **518**.

(24) The puck, as shown as puck **404** and puck **504**, can be a generally right cylinder-shaped object with opposing planar ends **532**, **534**, and a cylindrical side **522**. The slot **408**, **518** can pass from end to end along the bore, and extend from the bore outward to the cylindrical surface. The bore can run through the center of the puck, but it is also contemplated that there can be more than one bore through the puck so that multiple cables can be run through the same puck, with each bore having its own slot to allow a cable to be placed into the respective bore.

(25) FIG. 7 shows a cable reel housing **700** for use with a cable swipe cassette assembly, in accordance with some embodiments. The housing **700** is intended to hold a reel on which a signal cable is wound and that can be drawn out of the housing and off the reel, or back into the housing onto the reel. The reel is shown in FIG. 8. The housing **700** has a generally cylindrically shaped body **702** in the form of a cylinder having a length that is less than its diameter, and thus resembles more of a puck shape. The housing has two halves **710**, **712** that form opposing ends of the body **702**, and which interface around the outside of the body **702**, along the diameter of the body **702**. The body can be attached to a connecting boss **704** that allows the body **702** to be attached to, or otherwise mounted on a medical device stand. Further, the body **702** has a cable inlet **708** through which one end of the cable passes and connects to the medical device. The body **702** also has a cable outlet **706** through which the cable passes from the reel inside the housing to a sensor or medical component that is monitored or controlled by the medical device. As shown here, the cable outlet **706** does not allow for a cable swipe cassette assembly (e.g. **500**) to be mounted on the housing at the cable outlet **706**, but in some embodiments, there is a feature at the cable outlet **706** that allows for the cable swipe cassette assembly to be mounted onto the housing **702**. In cases where the cable swipe cassette assembly cannot be mounted directly on the reel housing **700**, a bracket or other suitable structure is preferably used to hold the cable swipe cassette assembly in front of the cable outlet **706**.

(26) FIG. 8 shows an exploded view of a cable reel housing **700** for use with a cable swipe cassette assembly, in accordance with some embodiments. In addition to the housing portions **710**, **712**, there is shown in housing portion **712** an axial spindle **810** on which a rotatable reel member **802** is mounted. The reel member **802** is able to revolve around the spindle **810**. The reel member **802** includes a spring basket **804** that is made of several wall members that extend from the radial wall **803** around a central opening in the radial wall **803** through which the spindle **810** passes. The wall members can follow a circle centered around the central opening. A coil spring **806** is housed

in the spring basket **804** and has an inner end of the coil attached to the spindle and the outer end of the coil attached to the reel member **802**, such as the spring basket **804**. Thus, when the reel member **802** is rotated, it loads the spring **806** and biases the spring to cause the reel member **802** to turn the opposite direction from that in which it is turned to load the spring **806**. A cover **808** is placed over the spring **806** to hold the spring **806** in the basket **804**, and can latch to the wall members of the basket **804** to retain the cover **808** on the basket **804**. The spindle **810** can have a distal end that meet the radial wall of housing portion **710** to hold the reel member **802** between and within the two housing portions **710**, **712**.

(27) A cable can be inserted through an opening in the center of the radial wall of housing portion **712**, through the center of the reel member **802**, the spring, **806** and cover **808** using a swivel connector that allows one portion of the cable to rotate relative to the other portion opposite the swivel connector. Alternatively, the swivel connector can be built into the reel at the center of the housing portion **712**. A cable portion is wound around the basket **806**, having an end fed through the cable outlet **706**. Thus, when the cable is pulled through the cable outlet **706**, it causes the reel member **802** to turn, which loads the spring **806**. The reel **802** can be locked in place by a lever or similar interference structure, and once the cable is done being used, the lock can be removed, allowing the loaded spring to rotate the reel in the opposite direction and thereby drawing the cable back into the reel and around the basket **806**.

(28) FIG. **9** shows a medical device stand system **900** using several cable reels, each with a cable swipe cassette, in accordance with some embodiments. The system **900** includes a medical device **902** that has a graphic display. The medical device **902** is mounted on the top of a stand **904**, and specifically on the top of a central post of the stand **904**. The stand **904** can have several casters or wheels so that the stand **904** can be mobile and moved from patient to patient as needed. The medical device **902** can monitor several physiological parameters of a patient through electromotive sensors that are coupled to specific channels of the medical device through cable assemblies. The cable assemblies each include a reel **906** on which a wound portion **910** of a cable stored, and wound on a reel member inside the reel **906**. A static cable portion **908** is connected between the reel **906** and the medical device **902**. The static cable portion **908** can be connected to the wound portion **910** through a swivel connector that allows the wound portion to rotate about the swivel connector without twisting the cable. A cable swipe cassette assembly **912** is attached or mounted at the cable outlet of each reel. The cable swipe cassette assembly **912** can be substantially the same as cable swipe assembly **500**, and contains a puck that hold an antimicrobial fluid. As the cable **910** is moved through the puck in the cable swipe cassette assembly **912**, the antimicrobial fluid is imparted to the exterior surface of the cable, thereby disinfecting that portion of the cable's exterior surface. Thus, when the medical device is used with a new patient, each cable for each reel can be drawn out and attached to a sensor or electrode, and when pulled out of the reel, the exterior surface of the cable is disinfected. This greatly reduces the chances of transmitting or carrying infectious agents to the patient or the patient's garments or bedding. And when the medical device **902** is no longer needed, the cables are allowed to reel back up, passing again through the puck and being coated with the antimicrobial material. Thus, for most patients, the cable **910** is disinfected twice; once when it is drawn back onto the reel after use with a prior patient, and then again when pulled out of the reel.

(29) FIG. **10** shows a side view of a cable reel **906** and a clamp on cable swipe cassette **912**, separated, and FIG. **11** shows a side view of the cable reel **906** and a clamp on cable swipe cassette **912**, with the cable swipe cassette **912** clamped onto the cable reel, in accordance with some embodiments. In both figures the cable swipe cassette **912** is shown in a closed configuration. In FIG. **10** it can be seen that the cable reel **906** has a cable outlet **1002** that includes one or more annular ridges or shoulders **1003** around the cable outlet **1002**. The cable swipe cassette **912** includes, on one side, an annular wall **1006** that extends outward from the side of the cable swipe cassette **912**. Half of the annular wall is on the upper portion of the housing of the cable swipe

cassette **912**, and half is on the bottom portion of the housing of the cable swipe cassette **912**. By opening the housing, as is shown for example in FIG. 5, the upper and lower portions of the annular wall **1006** come apart. Inside the annular wall **1006** is an annular ridge **1008** that is configured to fit over a first shoulder **1003** on the cable outlet **1002** of the cable reel **906** at location **1004**. When the housing is then closed, the annular ridge **1008** is captured on the other side of shoulder **1003**, creating a mechanical interference that prevents the housing from coming off the cable outlet **1002**. FIG. 11 shows the cable swipe cassette **912** clamped onto the cable outlet **1002** with the reel cable portion **910** passing through the cable swipe cassette. The reel **906** can have feet **1010** or some other structure for connecting to a tray or other structure on the stand **904** (FIG. 9).

(30) FIG. 12 shows a series of several cable swipes that all fit in the same cassette housing but which accommodate cables of different sizes, in accordance with some embodiments. There are three pucks **1202**, **1204**, **1206** shown. Each has the same outer diameter **1208**, but the diameters of the bore through each of them, where the cable passes, can be different. Thus, the bore of puck **1202** has the smallest diameter **D1**, the bore of puck **1204** has a diameter **D2** that is larger than **D1**, and the bore of puck **1206** has a diameter **D3** that is larger still than **D2**. This allows different sized cables to be used with the same cable swipe cassette housing, and ensuring that the contact between the puck and the cable is neither too light or too heavy. If the contact is too light, then there may not be adequate distribution of the antimicrobial material to the cable's exterior surface. If the contact is too heavy, e.g. the bore is too small, then the drag on the cable between the material of the puck and cable's exterior surface may prevent the reel from drawing the cable back into the reel.

(31) A cable wetting system has been disclosed in which a cable swipe cassette holds a puck that is made of a porous resilient material and there is a fluid disposed in the puck. A cable is placed into the cable swipe cassette so that the cable runs through bore through the puck. The puck is held in place, such as by a cable swipe cassette housing, so that the cable can be drawn through the puck, resulting in some of the fluid in the puck wetting the outer surface of the cable. In some embodiments the fluid can be an antimicrobial fluid, and the cable can be a cable for a medical device that is used to connect the medical device to a medical or physiological sensor on a patient. In some embodiments a portion of the cable can be stored on a reel, and drawn from the reel and through the puck when pulling the cable out for use. In some embodiments it is contemplated that the cable can be stored loosely, and drawn through the puck as needed.

Claims

1. A cable swipe cassette, comprising: a puck made of a porous resilient material, the puck having a bore through the puck from a first side to a second side opposite the first side; and a housing having an upper portion and a lower portion, each of the upper portion and the lower portion having a first side and a second side, wherein the housing is configured to enclose the puck therein between the first side and the second side of the upper portion and between the first side and the second side of the lower portion, and wherein an opening is formed at the first side of the upper portion and the lower portion that correspond with the bore through the puck at the first side of the puck, and an opposing opening is formed at the second side of the upper portion and the lower portion that corresponds with the bore through the puck at the second side of the puck, and wherein the housing is sized such that the puck is compressed by the housing when the housing is closed around the puck.
2. The cable swipe cassette of claim 1, wherein the upper portion and the lower portion are coupled together at a hinge.
3. The cable swipe cassette of claim 2, wherein the housing has a latch opposite the hinge to latch the upper and lower portions of the housing together.
4. The cable swipe cassette of claim 1, further comprising a fluid disposed in the puck.

5. The cable swipe cassette of claim 4, wherein the fluid is an antimicrobial fluid.
 6. The cable swipe cassette of claim 1, wherein the puck further comprises a slot extending outward from the bore to a cylindrical surface of the puck.
 7. The cable swipe cassette of claim 5, wherein the bore is configured to wet an external surface of a cable being drawn through the bore.
 8. A disinfecting cable swipe cassette assembly for disinfecting an exterior surface of a cable, comprising: a puck made of a fluid absorbent material, the puck having a bore through the puck from a first side to a second side opposite the first side, and wherein a diameter of the bore is selected to be smaller than a diameter of the cable; and a housing having an upper portion and a lower portion, each of the upper portion and the lower portion having a first side and a second side, wherein the housing is configured to enclose the puck therein between the first side and the second side of the upper portion and between the first side and the second side of the lower portion, and wherein an opening is formed at the first side of the upper portion and the lower portion that correspond with the bore through the puck at the first side of the puck, and an opposing opening is formed at the second side of the upper portion and the lower portion that corresponds with the bore through the puck at the second side of the puck, and wherein the housing is sized such that the puck is compressed by the housing when the housing is closed.
 9. The disinfecting cable swipe cassette assembly of claim 8, wherein the puck is made of a porous resilient material.
 10. The disinfecting cable swipe cassette assembly of claim 8, further comprising the puck having a slot extending through the puck from the bore to an outside of the puck.
 11. The disinfecting cable swipe cassette assembly of claim 8, further comprising a fluid disposed in the fluid absorbent material.
 12. The disinfecting cable swipe cassette assembly of claim 11, wherein the fluid is an anti-microbial fluid.
 13. The disinfecting cable swipe cassette assembly of claim 8, further comprising a hinge connecting the upper portion to the lower portion at a side of the housing between the first sides and the second sides of the upper and lower portions of the housing, and wherein the housing is made of a fluid impermeable material.
 14. The disinfecting cable swipe cassette assembly of claim 13, further comprising a latch formed on the outside of the housing opposite the hinge.
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